

2023

COMPUTER SCIENCE AND ENGINEERING

Paper : CSEL-0902

Elective - I : Image Processing

Full Marks : 70

*The figures in the margin indicate full marks.**Candidates are required to give their answers in their own words as far as practicable.*Answer **question no. 1, question no. 2** and **any four** from the rest.1. Answer **any five** questions from the following :

2×5

(a) Perform shear (vertical) and shear (horizontal) transformation on the following matrix :

5	3	4	2
2	4	6	1
1	3	1	3
2	6	5	3

- (b) If you want to transform a selected area to be set to black, which transformation is suitable and why?
- (c) In arithmetic coding, what is the impact of an increase in the length of the sequence?
- (d) Calculate the aspect ratio of an image of dimension 640×480 .
- (e) How many different shades of greys are there in a colour RGB system with 12 bit RGB image? Justify your answer.
- (f) How does correlation differ from convolution? Illustrate with an example.
- (g) What is the impact of image multiplication in a region of interest?

2. Answer **any five** questions from the following :

4×5

- (a) There are some cases where the local histogram processing is better than the global histogram processing. Justify.
- (b) How can gradient operators be used for Edge Detection? Explain with an example.
- (c) Find the time required (in seconds) for transmitting a monochrome image of size 2.5 inch $\times$ 2 inch scanned at 150 DPI and to be sent at 28 kbps speed.
- (d) Quantization has a direct effect on spatial resolution. Justify with an example.

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- (e) Compute the median value of the masked pixels shown below using a 3×3 mask.

18	22	33	25	32	24
34	128	24	17	26	33
22	19	32	31	28	26

- (f) Suppose the RGB colour triplet for a particular colour is given by (0.3, 0.5, 0.2). Compute corresponding YIQ and HSV triplets.
- (g) "Bicubic interpolation yields sharper results than bilinear interpolation." Justify the statement.
3. (a) Out of different types of data redundancy, which one is important and why?
- (b) Find out the Huffman encoding and the code length of a message containing the following characters and frequency. Show the steps of source reduction and draw the Huffman tree.

Character	Frequency
a	2
b	8
c	9
d	17
e	25
f	50

4+6

4. (a) Discuss the process of region splitting and merging for region-based segmentation.
- (b) Cluster the dataset $\{(0, 0)', (0, 1)', (5, 4)', (5, 5)', (4, 5)', (1, 0)'\}$ by applying the K-means algorithm. Show the intermediate steps. (Assume $K = 2$)
- (c) What is the performance index used in the K-means algorithm? What are the problems of the K-means algorithm? 3+4+(1+2)
5. How can you define the 4×4 unitary DFT matrix F ? Using F , find the Fourier transform of the matrix $U = [1, 1, 1, 1]^T$. 10
6. (a) Write down the conditions for two pixels to be adjacent. Define 8-adjacency and m-adjacency.
- (b) Consider the following :

	3	4	1	2	0	$\leftarrow q$
	0	1	0	4	2	
	2	2	3	1	4	
$P \rightarrow$	3	0	4	2	1	

For $V = \{2, 3, 4\}$, find 8-path and m-path. Are they unique? If not, find the shortest one. 4+6

7. (a) The following decision functions are given for a 3-class problem :

$$d_1(x) = -x_1$$

$$d_2(x) = x_1 + x_2 - 1$$

$$d_3(x) = x_1 - x_2 - 1$$

- (i) Assume that these functions are determined under multiclass case 1 (where each class is separable from the other by a decision surface). Sketch the decision boundaries and regions for each pattern class.
- (ii) Assuming the case 2 condition where each class is separable from every other by a distinct decision surface. Suppose $d_{12}(x) = d_1(x)$, $d_{13}(x) = d_2(x)$, $d_{23}(x) = d_3(x)$, sketch the decision boundaries and regions for multiclass case 2.
- (b) Construct a PDL grammar using defined primitives and functions, which can generate numerals 0 and 5. (3+3)+4
8. Suppose A_i is the grey level of an input image which has to be transformed to output image gray level B_i using histogram equalization technique. For an 8-level image of size 64×64 , use the following frequency table for the input grey levels.

Gray level (A_i)	0	1	2	3	4	5	6	7
Frequency	123	78	281	417	639	1054	816	688

- (a) Find the frequency levels for the output gray level (I).
- (b) Prove with the help of histogram specification that desired I is attained. 5+5